

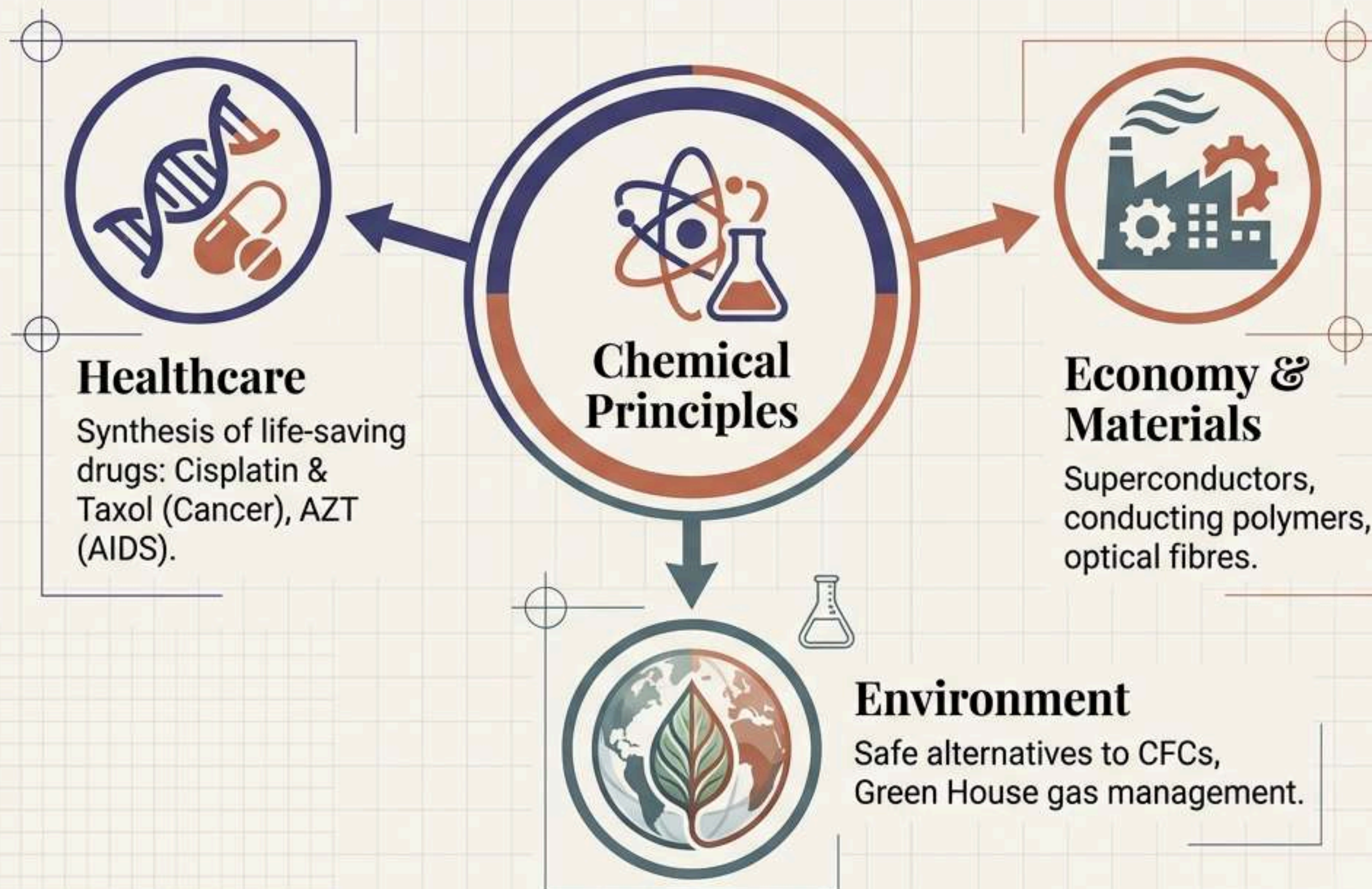
The Science of Matter

From Ancient Wisdom to Modern Measurement

Science is the continuing human effort to systematise knowledge for describing and understanding nature. Chemistry is the branch dedicated to the preparation, properties, structure, and reactions of material substances.

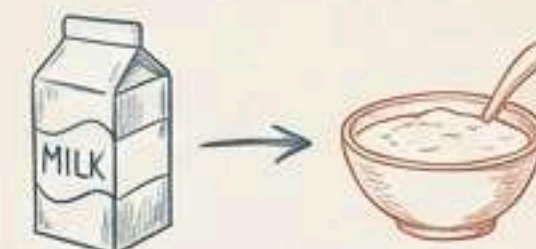
“Chemistry is the science of molecules and their transformations. It is the science not so much of the one hundred elements but of the infinite variety of molecules that may be built from them.” — Roald Hoffmann

Chemistry is the Backbone of Civilisation

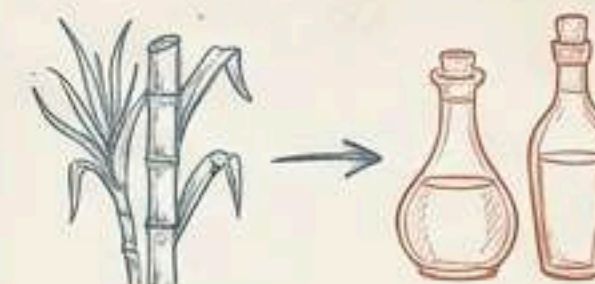


Daily Life

Chemistry explains the



see every day, from the curd formation in milk and vinegar from sugarcane to the rusting of iron.



Rusted iron chain.



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The Indian Legacy: Rasayan Shastra

Ancient Terminology

Early chemistry in India was known as Rasayan Shastra, Rastantra, or Rasvidya.

Metallurgy & Glass

Extraction of gold, silver, and copper. Hardness of copper artefacts improved using tin and arsenic.

'Faience' glass used in ornaments (1000–900 BCE).



Excavation Evidence

Mohenjodaro & Harappa showcased mass production of pottery, baked bricks, and gypsum cement containing lime and calcium carbonyl carbonate.

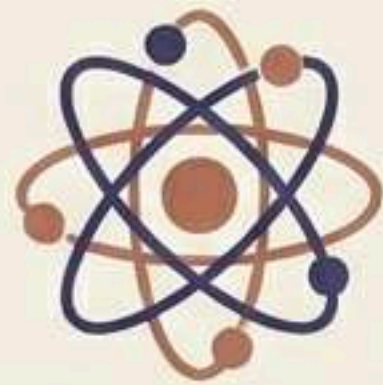
Chemical Arts

Tanning of leather and dyeing of cotton (Rigveda, 1000–400 BCE).
Glass colouring using metal oxides.



Philosophers of the Atom

600 BCE



Acharya Kanda (Kashyap)

Proposed the 'Paramanu'. Stated that all substances are aggregates of spherical, indestructible, eternal particles in motion. Conceptualised 2,500 years before Dalton.

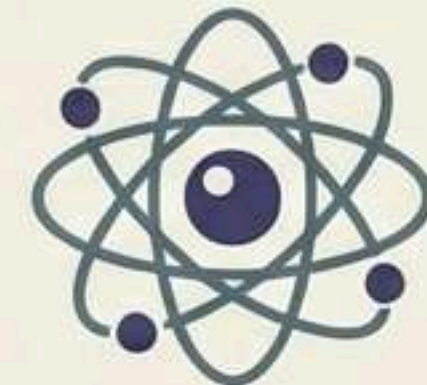
Ancient Era



Advanced Application

Nagarjuna (Rasratnakar) on mercury compounds. Charaka Samhita discusses 'Bhasma'—reduction of metal particle size, an early form of nanotechnology.

1800s



Modern Theory

John Dalton formalizes atomic theory in the West.



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The Physical Nature of Matter

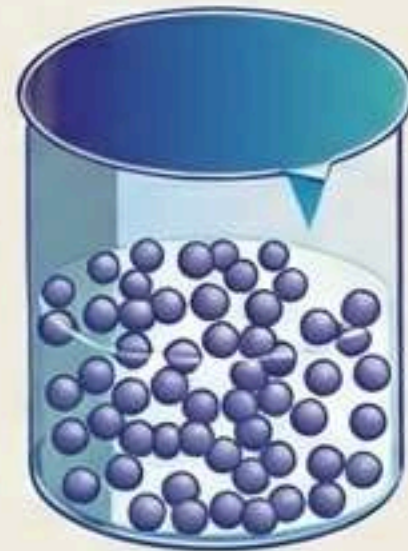
Matter is anything that has mass and occupies space.

Solid



Definite volume and shape.
Particles held in orderly
fashion with little freedom
of movement.

Liquid



Definite volume but no
definite shape. Particles
are close but can move
around.

Gas



No definite volume or shape.
Particles are far apart and
move easily and fast.

Interconvertibility via Heat & Pressure

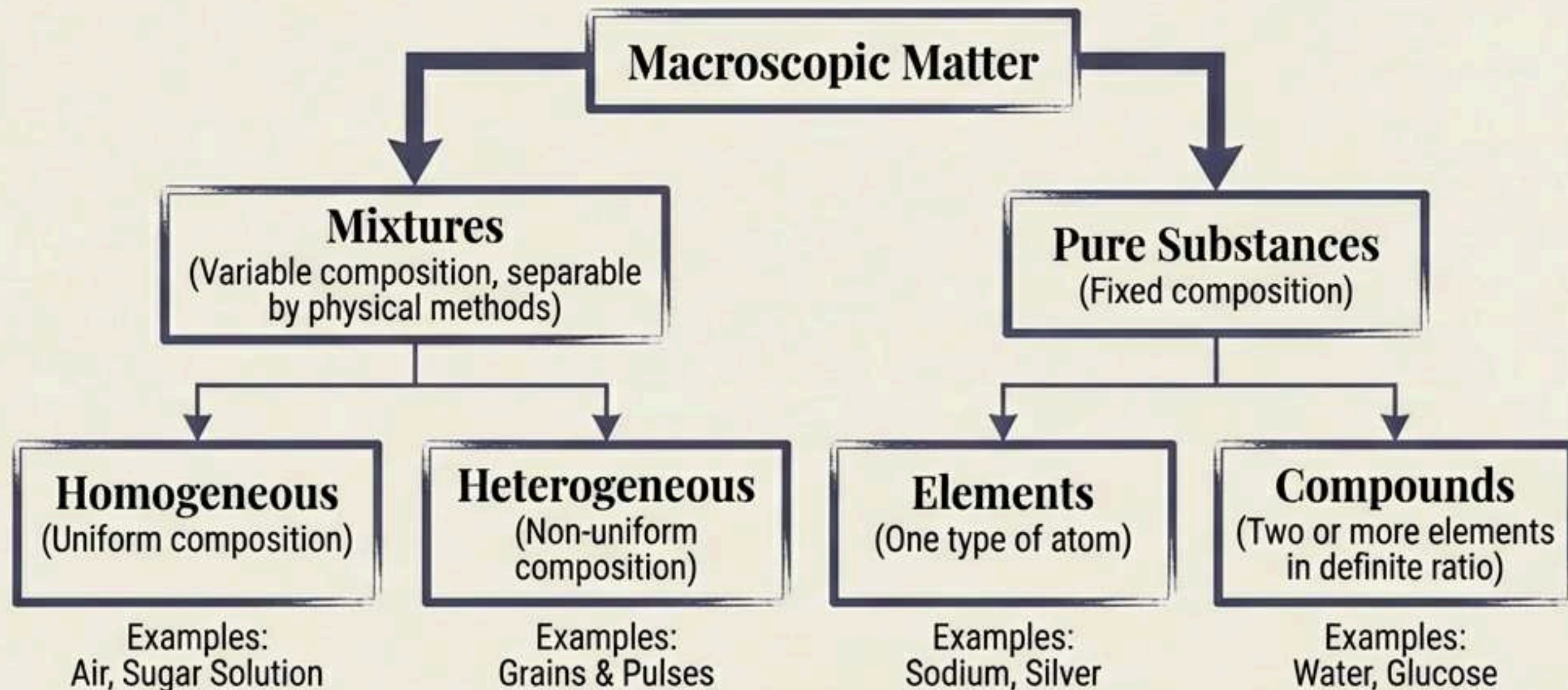


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Classifying the Material World

The physical and chemical categorization of matter.



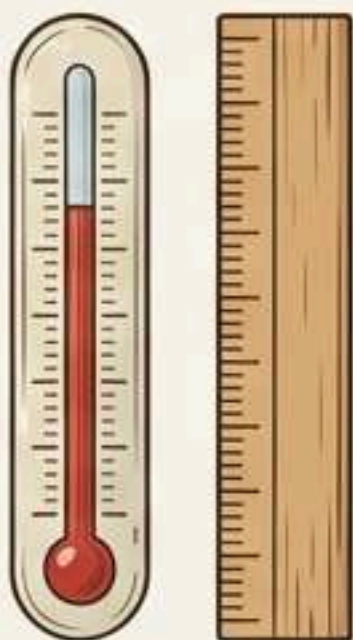
Key Insight

Constituents lose original properties.
(e.g., Hydrogen [burns] + Oxygen [supports combustion] → Water [fire extinguisher]).



Properties and Measurement

Physical Properties



Observable without changing identity or composition.

Colour, odour, melting point, boiling point, density

Chemical Properties



Observation requires a chemical change.

Acidity, basicity, combustibility

The Need for Standards

Quantitative measurement is represented by a number followed by a unit. Historically, systems varied (English vs Metric), leading to the need for a unified global standard.

The Universal Language: SI Units

Established by the CGPM (1960). Maintained in India by the National Physical Laboratory.

Quantity	Symbol	Unit Name	Unit Symbol
Length	l	Metre	m
Mass	m	Kilogram	kg
Time	t	Second	s
Electric Current	I	Ampere	A
Thermodynamic Temperature	T	Kelvin	K
Amount of Substance	n	Mole	mol
Luminous Intensity	I _v	Candela	cd

Derived Units: Mass, Volume and Density

Mass vs Weight

- **Mass:** Amount of matter (constant).
- **Weight:** Force exerted by gravity (varies by location).

Volume & Density

$$1 \text{ L} = 1000 \text{ mL} = 1000 \text{ cm}^3 = 1 \text{ dm}^3$$

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$$

Density indicates how closely particles are packed.

Temperature

Kelvin (K) is the SI unit.

$$K = ^\circ C + 273.15$$

Negative values are possible in Celsius, but not in Kelvin.



Scientific Notation

$1.66 \times 10^{-24} \text{ g}$
Laboratory Slate Inter

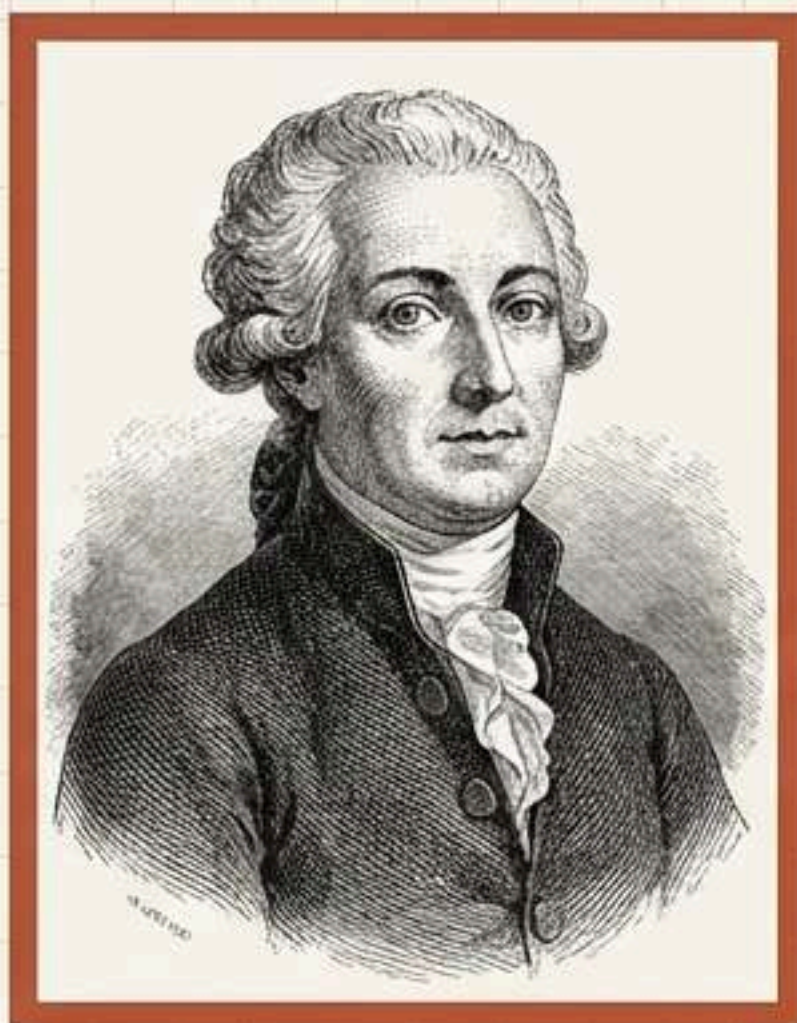
Meaningful digits known with certainty plus one estimated digit. Rule: **All non-zero digits are significant. Zeros between non-zeros are significant.**

A target with concentric rings of orange, white, and blue. Four arrows are shown hitting the center bullseye.

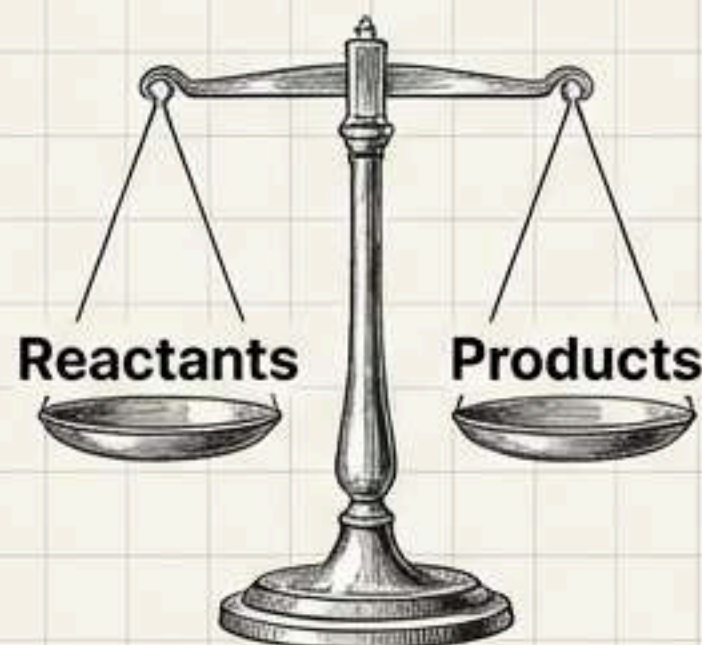
Precise & Accurate

Laws of Chemical Combination (I)

Law of Conservation of Mass (Lavoisier, 1789)



Matter can neither be created nor destroyed.



Law of Definite Proportions (Joseph Proust)

A given compound always contains exactly the same proportion of elements by weight.

Example:
Cupric Carbonate contains identical ratios of Copper, Carbon, and Oxygen whether natural or synthesized.



Laws of Chemical Combination (II)

Law of Multiple Proportions (Dalton, 1803)

If two elements form more than one compound, the masses of one element that combine with a fixed mass of the other are in the ratio of small whole numbers.

Example: Hydrogen + Oxygen $\rightarrow$ Water (16g O) vs. Hydrogen Peroxide (32g O). Ratio 1:2.

Gay Lussac's Law of Gaseous Volumes (1808)

Gases combine or are produced in a simple ratio by volume.



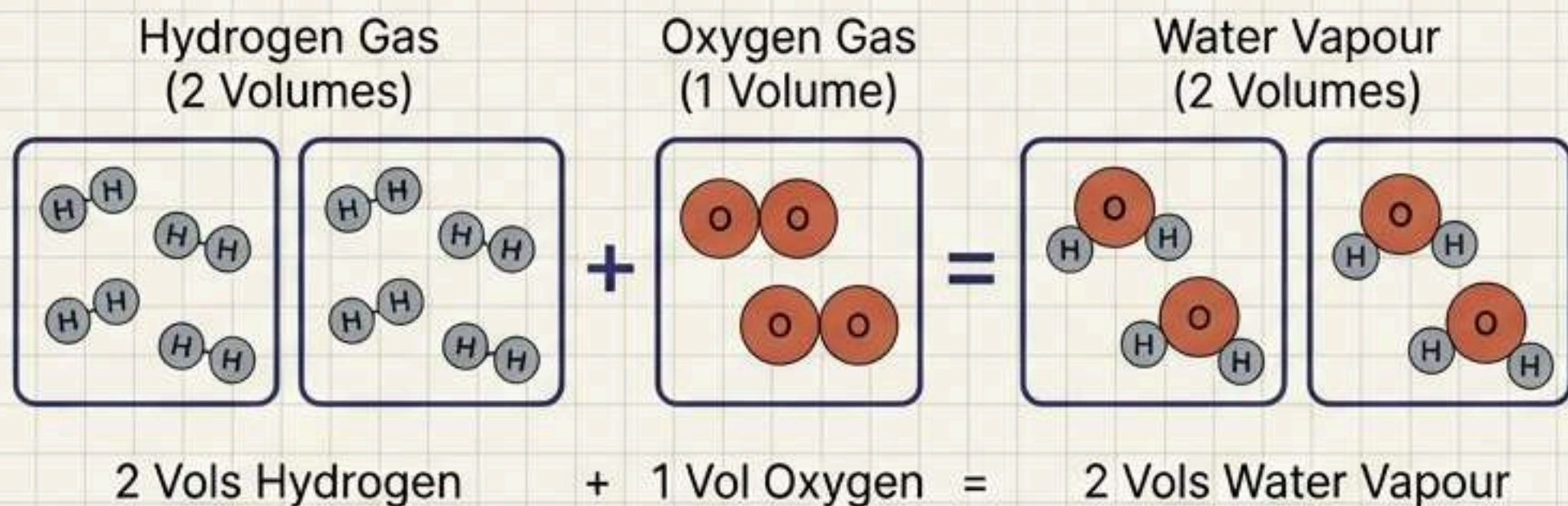
Avogadro's Insight (1811)



Amedeo Avogadro

Proposal: Equal volumes of all gases at the same temperature and pressure contain equal numbers of molecules.

Crucial Distinction: Avogadro distinguished between atoms and molecules.



Dalton's Atomic Theory (1808)

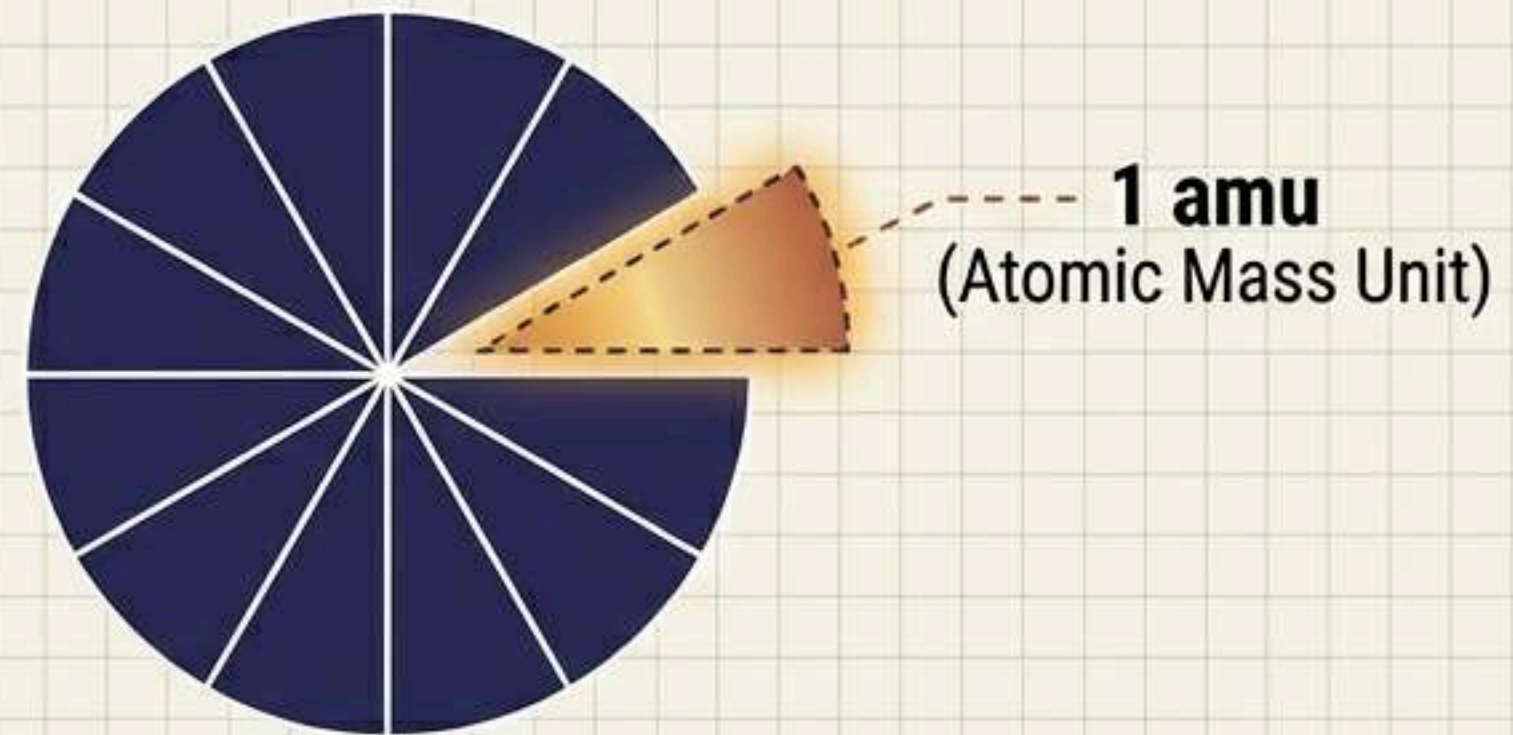
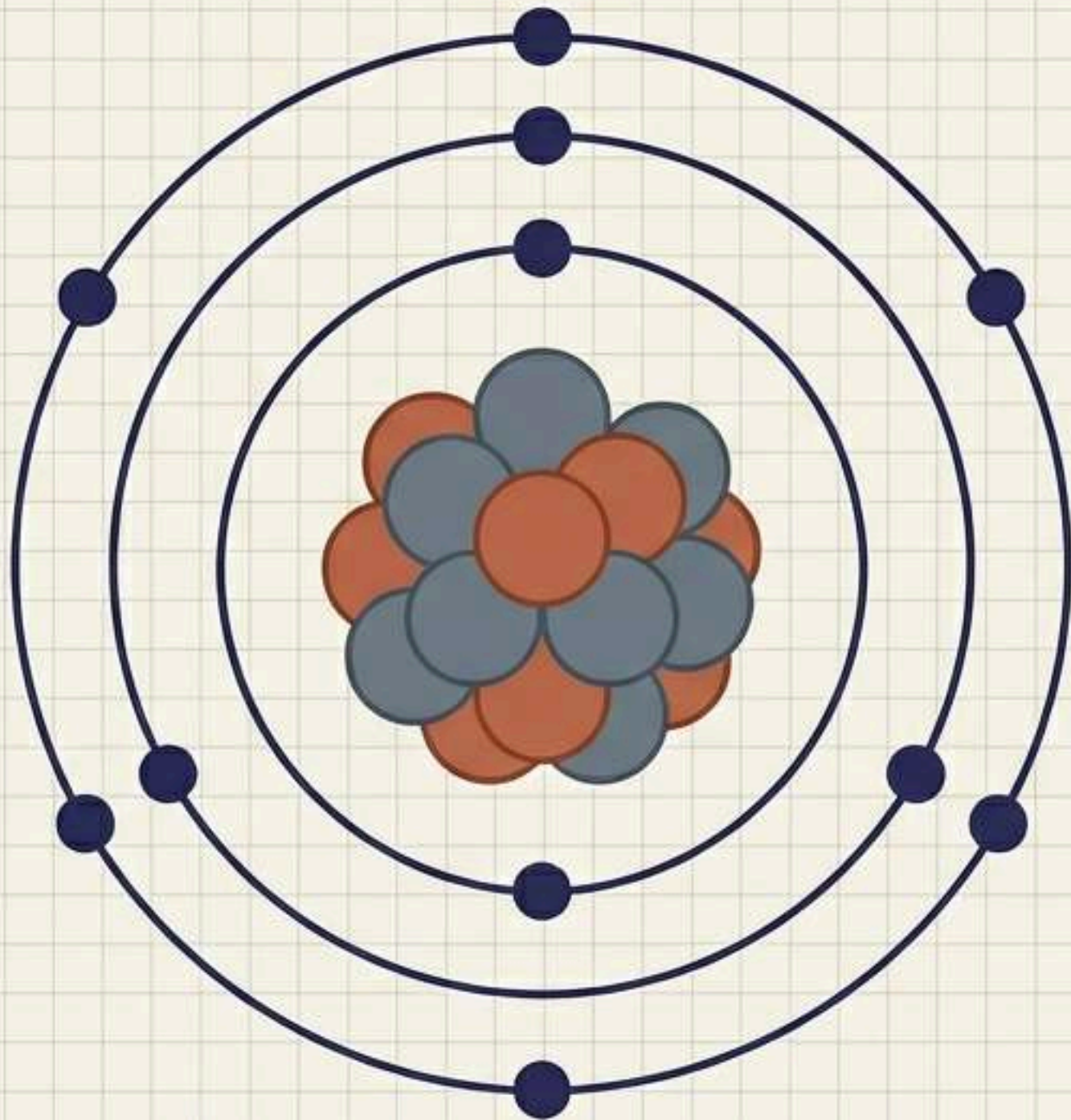
Source: *A New System of Chemical Philosophy*



John Dalton

- Matter consists of indivisible atoms.
- All atoms of a given element have identical properties and mass. Atoms of different elements differ in mass.
- Compounds are formed when atoms of different elements combine in a fixed ratio.
- Chemical reactions involve reorganisation of atoms; they are neither created nor destroyed.

Quantifying the Atom



The Standard (1961): Carbon-12 is the reference. One atomic mass unit (1 amu) is defined as a mass exactly equal to one-twelfth of the mass of one carbon-12 atom.

From the philosophical '*paramanu*' to the precise 'amu', we have successfully systematised the invisible building blocks of nature.